

Behavioural Features (Video Features)

“These tell us how the candidate appears and behaves during the interview.”

Answer: “We did not generate these features arbitrarily. Instead, we grounded them in published research on nonverbal behaviour and multimodal engagement in interviews. Studies consistently show that interviewers rely heavily on **visual and vocal cues** – for example, direct eye contact and confident body language – when evaluating candidates [25†L209-L217] [16†L135-L143]. Based on such insights, we defined features like **Engagement**, **Confidence**, **Hesitation**, and **Eye Contact**, and we generate them using logical rules inspired by the literature.

Step 1 – Literature Review

We reviewed research on **nonverbal communication in interviews** and multimodal affective sensing. Meta-analytic findings highlight that **eye contact** and body movements are among the strongest nonverbal predictors of interview success [25†L209-L217]. Video interview studies show that **gaze patterns** and micro-expressions reliably indicate candidate engagement and emotional state [16†L135-L143]. Similarly, human-computer interaction research demonstrates that **voice features** (pitch, tone, etc.) can predict user engagement in conversation [7†L65-L71]. These results confirm that cues like facial expression, speech tone, and eye contact carry meaningful information about a candidate’s performance, justifying our inclusion of related features.

Step 2 – Extract Behavioural Principles

From this literature, we distilled how each behavioural feature should relate to performance:

- **Engagement:** High attention and positive expressions correlate with engagement [16†L135-L143]. We link high engagement with high correctness.
- **Confidence:** A calm, steady voice and open facial expressions signal confidence; signs of stress (trembling voice, tense body) indicate lower confidence [28†L149-L152].
- **Hesitation:** Long pauses, filler words or stammering typically signal uncertainty or cognitive load, so hesitation increases when a candidate struggles.
- **Eye Contact:** Strong direct gaze usually conveys attentiveness and self-assurance [25†L209-L217].

Step 3 – Convert Principles into Data-Generation Rules

We translated these principles into quantitative rules for our synthetic data. For example:

- If a candidate’s semantic scores are very high (e.g. Correctness $\geq 90\%$), we assign **Engagement** = 0.80–0.95 and **Confidence** = 0.80–0.95, with **Hesitation** = 0.05–0.15 (low) [7†L65-L71].
- If semantic performance is low ($< 50\%$), we assign low engagement/confidence and high hesitation (e.g. Engagement 0.30–0.50, Confidence 0.20–0.40, Hesitation 0.60–0.90).
- **Eye Contact** is scaled similarly: high-correctness yields high eye-contact (e.g. 0.70–0.95), while poor performance gives low eye-contact (e.g. 0.30–0.60) [25†L209-L217].

These rules ensure the behavioural values are consistent with the candidate's apparent understanding and emotional state.

Step 4 – Generate Dataset

Applying these rules, we synthesize each interview turn into a dataset row:

Example

Correctness	Coverage	Reasoning	Missing	Engagement	Confidence	Hesitation	EyeContact	Policy
94	91	90	0	0.92	0.90	0.10	0.85	Increase Difficulty
71	68	74	5	0.55	0.50	0.60	0.40	Probe Missing Concept

In each row, a high correctness (94%) corresponds to high engagement/confidence and low hesitation, whereas a lower score (71%) yields only moderate engagement and higher hesitation. Each synthetic row thus ties behavioural signals to performance: strong answers come with high engagement and confidence and minimal hesitation, while weaker answers show reduced engagement and more hesitation.
